PADHAKU
Exploration · Class 9 Science



CHAPTER 8

Journey Inside the Atom

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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🤔 Think It Over

1 **Are atoms the smallest indivisible particles?**

✔ ANSWER

No. Atoms are made of even smaller particles: **electrons, protons and neutrons**.

Radioactivity and the discovery of electrons showed that atoms can be divided.

2 **Why do electrons not fall into the nucleus even though they are attracted to protons in it?**

✔ ANSWER

According to **Bohr's model**, electrons move only in fixed orbits (stationary states / shells).

While moving in a fixed shell, an electron **does not lose energy**, so it does not spiral into the nucleus.

3 **Why did scientists keep modifying atomic models?**

✔ ANSWER

Each new experiment gave new evidence that the old model could not explain.

For example, the gold foil experiment could not be explained by Thomson's model, and the stability of atoms could not be explained by Rutherford's model.

So models were changed and improved step by step.

⭐ Pause and Ponder · 8.2.1 Thomson's model of an atom

1 (i) **In your clay-and-beads Thomson 'atom', what will happen if the positive charge on the clay is less than the total negative charge of the beads?**

✔ ANSWER

The total negative charge would be more than the positive charge.

So the model would have an **overall negative charge**; it would not be a neutral atom.

1 (ii) **What if, by mistake, the clay itself carries a bit of negative charge? Would your model still represent a neutral atom?**

✔ ANSWER

No. The positive charge would become less, so the negative charge of the beads would be more.

The model would be **negatively charged**, not neutral.

2 **Could an orange or a lemon, which also contain seeds inside soft pulp, be a good comparison? In what ways does it match Thomson's idea and where does it fall short?**

💡 SAMPLE ANSWER

Matches: seeds (like electrons) are inside a soft pulp (like the sphere of positive charge).

Falls short: in an orange or lemon the seeds are mostly near the centre, not spread evenly through the whole fruit; the fruit also has a peel and segments, which Thomson's atom does not have.

So a watermelon is a better comparison.

3 **Why did Thomson conclude that electrons are present in all atoms?**

✔ ANSWER

The cathode rays were the same whatever the **material of the cathode** and whatever **gas** was filled in the tube.

So electrons must be a basic part of **all atoms**.

🤔 Think as a Scientist · 8.2.2 The gold foil experiment

? **Observe Fig. 8.4. Predict the observations you would expect if the gold foil in the experiment were made thicker. Also, draw a simple diagram to show the observations you expect.**

💡 SAMPLE ANSWER

A thicker foil has many more layers of atoms, so an α-particle meets **more nuclei** on its way.

- **Fewer** α-particles pass straight through.
- **More** α-particles are deflected, and more bounce back.
- Some α-particles may be stopped inside the foil.

Diagram: draw the beam hitting a thick foil, with only a few straight arrows going through and many arrows bending away or turning back.

⭐ Pause and Ponder · 8.2.2 Rutherford's model

4 **What do you think would happen if α-particles were replaced with negatively charged particles in**

Rutherford's gold foil experiment?

✔ ANSWER

Negatively charged particles would be **attracted** by the positive nucleus, not repelled.

Their paths would bend **towards** the nucleus; they would not bounce back as some α-particles did.

So the experiment would not show the result that proved a small positive nucleus.

5 **Rutherford found that a few α-particles bounced back sharply. How does this single surprising result completely rule out Thomson's 'plum pudding model' of the atom?**

✔ ANSWER

To push a fast, heavy α-particle back, there must be a **strong positive charge and large mass packed in a tiny place**.

In Thomson's model, the positive charge is spread evenly through the whole atom, so it is too weak at any point to turn an α-particle back.

So the bounce-back can be explained only by a small, dense, positive **nucleus**.

6 **If you could ask Rutherford one question about his work, what would it be?**

💡 SAMPLE ANSWER

"How did you decide that the electrons revolve around the nucleus, and what did you think keeps them from falling into it?"

Pause and Ponder · 8.2.2 C. Discovery of the proton

7 **Assertion (A): Rutherford concluded that most of the mass of an atom is concentrated in a small region at the centre called the nucleus.**

Reason (R): According to Thomson's model, electrons are embedded in a uniformly distributed positive charge sphere.

(i) Both A and R are true, and R is the correct explanation of A.

(ii) Both A and R are true, but R is not the correct explanation of A.

(iii) A is true, but R is false.

(iv) A is false, but R is true.

✔ ANSWER

(ii) Both A and R are true, but R is not the correct explanation of A.

A comes from the gold foil experiment; R only describes Thomson's (older) model.

8.3 What Components Contribute to the Mass of an Atom?

What if **What if an atom had no empty space? How would this have affected the size of various objects?**

💡 SAMPLE ANSWER

Almost all of an atom is empty space; the nucleus is about 1 lakh (10^5) times smaller than the atom.

Without empty space, all the matter would be packed as tightly as in a nucleus.

So every object would become **extremely tiny and extremely dense** (very heavy for its size).

⭐ Pause and Ponder · 8.4 Symbols of Elements

8 **Imagine you are a scientist who has discovered a new element. Name this element after yourself and justify that the symbol you have chosen follows the IUPAC rules.**

💡 SAMPLE ANSWER

Example: name **Ravi** → element **Ravium**, symbol **Rv**.

- First letter is capital (R), second letter is small (v).
- It is made from the first letter and another letter of the name.
- No other element already uses Rv, so it is unique.

9 **What problems could arise if every scientist used different symbols for the same element?**

✔ ANSWER

- Scientists in different countries would not understand each other's work.
- Formulae and equations would be confusing.
- Mistakes could happen in laboratories, medicines and industry.

A common symbol for each element avoids this.

⭐ 8.5 Atomic Number

? **Can you now say that elements with different atomic numbers are distinct from each other, and the atomic number uniquely identifies an element?**

✔ ANSWER

Yes. The atomic number (number of protons) is different for every element.

So each atomic number belongs to only one element.

? **Observe Fig. 8.10. How many neutrons and protons are present in a lithium atom, and what is its atomic number?**

✔ ANSWER

Lithium has **3 protons** and **4 neutrons**.

Atomic number **Z = 3**.

⭐ Pause and Ponder · 8.6 Mass Number

10 **An atom with an atomic number of 26 has 56 nucleons. Find out its number of electrons, protons and neutrons.**

✔ ANSWER

Protons = Z = **26**

Electrons = protons = **26**

Neutrons = A − Z = 56 − 26 = **30**

11 **The nucleus of an atom contains 20 protons. If its mass number is 41, find the number of neutrons in**

it.

✔ ANSWER

Neutrons = A − protons = 41 − 20 = **21**

12 **An atom has 18 neutrons and an atomic number of 17. What is its mass number?**

✔ ANSWER

A = protons + neutrons = 17 + 18 = **35**

13 **An atom ^{23}A has 11 electrons. Find the number of neutrons in it.**

✔ ANSWER

Protons = electrons = 11

Neutrons = 23 − 11 = **12**

⭐ 8.7 How Are Electrons Distributed in Different Energy Levels?

? **Helium contains two protons in its nucleus and two electrons. In which way will the two electrons be arranged in its atomic shell?**

✔ ANSWER

The K-shell can hold 2 electrons ($2 \times 1^2 = 2$).

So **both electrons are in the K-shell**: configuration **2**.

14 **Identify the number of electrons in the outermost shell of: (i) $^{12}_{6}C$ (ii) $^{19}_{9}F$ (iii) $^{28}_{14}Si$**

✔ ANSWER

(i) C: Z = 6 → 2, 4 → **4** electrons

(ii) F: Z = 9 → 2, 7 → **7** electrons

(iii) Si: Z = 14 → 2, 8, 4 → **4** electrons

15 **Write the electronic configuration of the elements having atomic numbers 12, 16 and 18.**

✔ ANSWER

Atomic number	Element	K	L	M
12	Magnesium (Mg)	2	8	2
16	Sulfur (S)	2	8	6
18	Argon (Ar)	2	8	8

So: **2, 8, 2**; **2, 8, 6**; **2, 8, 8**.

16 **Solve this riddle: I am an atom with a mass number of 23 and 11 protons. I am a soft metal and react vigorously with water. Who am I and how many neutrons do I have? You can also create one such riddle.**

💡 SAMPLE ANSWER

11 protons → atomic number 11 → **Sodium (Na)**.

Neutrons = 23 − 11 = **12**.

My riddle: "I have 8 protons and 8 neutrons. You breathe me in every moment. Who am I?" → **Oxygen** (mass number 16).

⭐ 8.8 Combining Capacity of an Atom: Valency

? **In NH_3 (ammonia) and $MgCl_2$ (magnesium chloride), what will be the combining capacities of nitrogen and magnesium respectively?**

✔ ANSWER

N combines with 3 H atoms → combining capacity of nitrogen = **3**.

Mg combines with 2 Cl atoms → combining capacity of magnesium = **2**.

? **Can you predict what happens to the atoms that already have eight electrons in their outermost shell (except elements with one shell only, where only two are possible)? Will they still try to lose or gain electrons?**

✔ ANSWER

No. Their octet is already complete, so they are stable.

They do not lose, gain or share electrons easily; they are largely unreactive and their valency is **0** (e.g. neon, argon).

? **Examine Table 8.4. Add one more column to it, and write down the common valency of each element.**

✔ ANSWER

Element	Distribution	Valency
H	1	1
He	2	0
Li	2, 1	1

Be	2, 2	2
B	2, 3	3
C	2, 4	4
N	2, 5	3
O	2, 6	2
F	2, 7	1
Ne	2, 8	0
Na	2, 8, 1	1
Mg	2, 8, 2	2
Al	2, 8, 3	3
Si	2, 8, 4	4
P	2, 8, 5	3
S	2, 8, 6	2
Cl	2, 8, 7	1
Ar	2, 8, 8	0

Rule: 1–3 valence electrons → valency = number of valence electrons; 5–7 → valency = 8 − valence electrons; 4 → 4.

? **All atoms of an element have the same number of electrons and protons. Can you say the same regarding the number of neutrons too?**

✔ ANSWER

No. Atoms of the same element can have different numbers of neutrons. Such atoms are called **isotopes**.

8.9.1 Isotopes

? **Protium ($^{1}_{1}H$), deuterium ($^{2}_{1}H$) and tritium ($^{3}_{1}H$): can you guess how many electrons each of these isotopes have?**

✔ ANSWER

Each has 1 proton, so each neutral atom has **1 electron**.

Pause and Ponder · 8.9.1 A. Average atomic mass

17 **Two different atoms have 11 protons each, but one has 12 neutrons, and the other has 13 neutrons. How do their atomic numbers and mass numbers compare? Are they the same element or different elements?**

✔ ANSWER

Atomic numbers: both **11** (same).

Mass numbers: 11 + 12 = **23** and 11 + 13 = **24** (different).

They are the **same element (sodium)**; they are **isotopes**.

18 **If a bromine atom is available in the form of two isotopes, $^{79}_{35}Br$ (49.7%) and $^{81}_{35}Br$ (50.3%), calculate the average atomic mass of the bromine atom.**

✔ ANSWER

Average atomic mass = 79 × 49.7/100 + 81 × 50.3/100

= 39.263 + 40.743

= 80.006 u

≈ 80 u

⭐ Revise, Reflect, Refine

1 **Choose the correct options and explain the reason for the correct and incorrect options in the context of Rutherford's gold foil experiment:**

(i) The experiment clearly showed the existence of neutrons in the nucleus.

(ii) The results disproved the plum pudding model and led to the idea of a nucleus at the centre of the atom.

(iii) The large deflection of a few alpha particles indicated that most of the mass of the atom and positive charge are packed into a tiny centre.

(iv) The way alpha particles were deflected showed that electrons move around the nucleus.

✔ ANSWER

Correct: (ii) and (iii).

(i) Incorrect: the experiment showed only a positive nucleus; neutrons were discovered later (1932) by James Chadwick.

(ii) Correct: a few α-particles bounced back, which the plum pudding model could not explain, so Rutherford proposed a central nucleus.

(iii) Correct: only a tiny, dense, positive centre can deflect α-particles through large angles.

(iv) Incorrect: α-particles were deflected by the nucleus, not by electrons; the idea that electrons revolve around the nucleus was Rutherford's proposal, not something the deflections showed.

2 **Which of the following statements are correct or incorrect according to Bohr's atomic model? Give a reason for each.**

(i) Electrons lose energy while moving in fixed orbits and slowly fall into the nucleus.

(ii) Electrons can exist anywhere around the nucleus with no fixed energy.

(iii) Electrons revolve around the nucleus in orbits of fixed energy without losing energy.

(iv) Electrons can be found between energy levels as they move around the nucleus.

✔ ANSWER

(i) **Incorrect:** in a fixed orbit (stationary state) an electron does not lose energy.

(ii) **Incorrect:** electrons move only in fixed shells, each with a definite energy.

(iii) **Correct:** this is Bohr's main idea; it explains why atoms are stable.

(iv) **Incorrect:** electrons can be only in the allowed shells, not in between them.

3 **Composition of the nuclei of three atomic species:**

Particle	X	Y	Z
Protons	18	17	17
Neutrons	19	18	20

Explain the relation between (i) Y and Z (ii) Z and X.

✔ ANSWER

Mass numbers: X = 18 + 19 = 37; Y = 17 + 18 = 35; Z = 17 + 20 = 37.

(i) Y and Z: same atomic number (17), different mass numbers (35, 37) → **isotopes** (of chlorine).

(ii) Z and X: same mass number (37), different atomic numbers (17, 18) → **isobars**.

4 **What conclusion did Rutherford draw about the position and characteristics of the atom's positively charged part based on the few alpha particles that bounced back or were deflected at large angles?**

✔ ANSWER

- The positive charge is **not spread out**; it is concentrated at the **centre** of the atom in the **nucleus**.
- The nucleus is **very small** (about 10^5 times smaller than the atom) and **dense**.
- It holds all the positive charge and most of the mass of the atom.

5 **Explain and arrange in the correct chronological order:**

(i) Bohr's model: electrons move in fixed orbits, each with a definite energy.

(ii) Thomson's model: 'plum pudding', electrons in a sphere of positive charge.

(iii) Rutherford's model: atoms have a dense central nucleus.

(iv) Dalton's model: atoms are indivisible particles.

✔ ANSWER

(iv) → (ii) → (iii) → (i)

- Dalton (1808): atoms are indivisible.

- Thomson (after discovering electrons, 1897): electrons spread in a positive sphere.
- Rutherford (gold foil experiment, 1911): tiny dense positive nucleus, electrons revolve around it.
- Bohr (1913): electrons in fixed energy shells, which explains stability.

6 **Electrons move around the nucleus in orbits. Why do they not fly away from the atom? Explain what keeps them attracted to the nucleus.**

✔ ANSWER

Electrons are **negatively** charged and the nucleus is **positively** charged (protons).

Opposite charges attract, so this **electrostatic force of attraction** holds the electrons around the nucleus and keeps them from flying away.

7 **Assertion (A): The discovery of subatomic particles helped in understanding the atomic structure.**
Reason (R): The number of electrons is equal to the number of protons in an atom.
(i) Both true, R explains A. (ii) Both true, R does not explain A. (iii) A true, R false. (iv) A false, R true.

✔ ANSWER

(ii) Both A and R are true, but R is not the correct explanation of A.

R is a true fact (atoms are neutral), but it does not explain why discovering the particles helped us understand the atom.

8 **For an atom of magnesium with mass number 24 and atomic number 12, determine the number of (i) protons, (ii) neutrons, (iii) electrons, and illustrate the arrangement of electrons.**

✔ ANSWER

(i) Protons = Z = **12**

(ii) Neutrons = 24 − 12 = **12**

(iii) Electrons = **12**

Arrangement: K = 2, L = 8, M = 2 → **2, 8, 2**

Diagram: a nucleus (12p, 12n) at the centre with three circles: 2 dots on the first, 8 on the second, 2 on the third.

9 **Find for the elements in Fig. 8.17: (i) name (ii) symbol (iii) total electrons (iv) valence electrons (v) valency (vi) protons (vii) atomic number.**

(a) 2 shells: 2, 1 electrons (b) 2 shells: 2, 5 (c) 3 shells: 2, 8, 3 (d) 2 shells: 2, 7

✔ ANSWER

Item	(a)	(b)	(c)	(d)
(i) Name	Lithium	Nitrogen	Aluminium	Fluorine
(ii) Symbol	Li	N	Al	F
(iii) Total electrons	3	7	13	9
(iv) Valence electrons	1	5	3	7
(v) Valency	1	3	3	1
(vi) Protons	3	7	13	9
(vii) Atomic number	3	7	13	9

10 **Both Rutherford's and Bohr's models have electrons orbiting the nucleus. Why did Rutherford's model fail to explain atomic stability, while Bohr's model succeeded?**

✔ ANSWER

Rutherford: an electron moving in a circle is accelerating, so it should lose energy, spiral inward and fall into the nucleus. Then atoms would collapse, but they are stable.

Bohr: electrons move only in fixed shells (stationary states) where their **energy stays constant**. They do not lose energy, so they do not fall in, and the atom is stable.

11 **An atom ^{70}X has 31 electrons. How many neutrons are there in its nucleus?**

✔ ANSWER

Protons = electrons = 31

Neutrons = 70 − 31 = **39**

12 **An atom has 79 protons and a mass number of 197. Calculate (i) the number of neutrons, and (ii) the number of electrons.**

✔ ANSWER

(i) Neutrons = 197 − 79 = **118**

(ii) Electrons = protons = **79**

13 **Complete Table 8.5:**

Atomic no.	Mass no.	Neutrons	Protons	Electrons	Element

5	–	6	–	–	–
–	14	–	–	7	Nitrogen
–	24	–	12	–	–
15	–	16	–	–	–
–	1	0	–	–	–

✔ ANSWER

Atomic no.	Mass no.	Neutrons	Protons	Electrons	Element
5	**11**	6	**5**	**5**	**Boron**
7	14	**7**	**7**	7	Nitrogen
12	24	**12**	12	**12**	**Magnesium**
15	**31**	16	**15**	**15**	**Phosphorus**
1	1	0	**1**	**1**	**Hydrogen**

Use: protons = electrons = atomic number; mass number = protons + neutrons.

14 **Element X has mass number 35 and 18 neutrons. (i) How many electrons and protons does X have? (ii) Atomic number? (iii) Identify X. (iv) Electronic configuration? (v) Valence electrons? (vi) Mass number if two neutrons are added? (vii) Relation of X with the new atom?**

✔ ANSWER

(i) Protons = 35 − 18 = **17**; electrons = **17**

(ii) Atomic number = **17**

(iii) X is **chlorine (Cl)**

(iv) **2, 8, 7**

(v) **7** valence electrons

(vi) 35 + 2 = **37**

(vii) Same atomic number, different mass numbers → they are **isotopes** (^{35}Cl and ^{37}Cl).

15 **An atom has 12 protons and 12 neutrons. All its electrons are replaced by hypothetical particles with the same charge as electrons but 500 times heavier. What effect will this have on its (i) atomic number (ii) atomic mass (iii) mass number (iv) overall charge?**

✔ ANSWER

(i) Atomic number: **no change**, still 12 (it depends only on protons).

(ii) Atomic mass: **increases**. Electrons' mass was negligible, but particles 500 times heavier add noticeable mass (an electron is about 1/1836 of a proton, so 12 new particles add about 12 × 500/1836 ≈ 3 u).

(iii) Mass number: **no change**, still 24 (it counts only protons + neutrons).

(iv) Overall charge: **no change**, still neutral (12 positive + 12 negative = 0).

⭐ The Quest Continues

? **Is it possible to completely understand everything that happens inside an atom?**

💡 SAMPLE ANSWER

Not yet. Each model (Dalton → Thomson → Rutherford → Bohr → quantum model) explained more, but new questions keep coming.

> Today we can only predict where electrons are most likely to be, not exactly where they are. Science keeps improving our understanding through new experiments.

🎉 **Chapter 8 done! Revise the answers once more tomorrow.**